

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                       | Additional Guidance | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------|
| 1(a)(i)         | <p>An explanation that includes 3 of the following points:</p> <ul style="list-style-type: none"> <li>• it {carries / transports} the male {gametes / nuclei / generative nucleus} (1)</li> <li>• to the ovule / micropyle / ovary / ovum / female gamete (1)</li> <li>• to {fuse with / fertilise} {ovum / egg (cell) / female gamete / female nucleus / polar nuclei} (1)</li> <li>• digest tissue of style (1)</li> </ul> | Accept sperm        | (3)  |

| Question Number | Answer                                                                                                                                                                                                                                                                             | Additional Guidance | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------|
| 1(a)(ii)        | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• the tubes grow towards the {micropyle / ovule / ovary / ovum / female gamete} (1)</li> <li>• {due to chemicals (released by the embryo sac) / it is chemotropic} (1)</li> </ul> |                     | (2)  |

| Question Number | Answer                                                                                                                                                                              | Additional Guidance | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------|
| 1(b)(i)         | <p>An answer that includes the following:</p> <ul style="list-style-type: none"> <li>• percentage germination / number germinating (1)</li> <li>• pollen tube length (1)</li> </ul> |                     | (2)  |

| Question Number | Answer                                                                                                                                                                                                                                               | Additional Guidance               | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------|
| 1(b)(ii)        | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• it involves {enzymes / (chemical) reactions} (1)</li> <li>• example of how {enzymes / (chemical) reactions} are affected by temperature (1)</li> </ul> | e.g kinetic energy / denaturation | (2)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                             | Additional Guidance                                                                                                                                                                                                   | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(c) (i)        | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• description of method (1)</li> <li>• calculation of dilution factor / use of <math>C_1V_1=C_2V_2</math> to calculate sucrose solution volume as 2 (cm<sup>3</sup>) (1)</li> <li>• stating the volume of sucrose solution and calcium ion solution to be used to make 10cm<sup>3</sup> (1)</li> </ul> | <p>ACCEPT mixed / dilute</p> <p>5x / by 5/ 1:4 / 1 in 5</p> <p>e. g. 2 cm<sup>3</sup> of sucrose solution to 8 cm<sup>3</sup> calcium ion solution / allow removal of 10cm<sup>3</sup> of bigger volume allow ecf</p> | (3)  |

| Question Number | Answer                                                                                                    | Additional Guidance             | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------|---------------------------------|------|
| 1(c)(ii)        | <ul style="list-style-type: none"> <li>varying the (concentration of) calcium ion solution (1)</li> </ul> | ACCEPT use of buffer (solution) | (1)  |

| Question Number | Answer                                                                                                                                                                                                                                                              | Additional Guidance                                        |                 |  | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------|--|------|
| 1(d)(i)         | <p>An answer including the following:</p> <ul style="list-style-type: none"><li>• suitable table drawn (1)</li><li>• all headings correct with units (1)</li><li>• all data for germination (only) and the 5 sucrose concentrations entered correctly (1)</li></ul> | Sucrose concentration / solution / mol dm <sup>-3</sup>    | Germination (%) |  | (3)  |
|                 |                                                                                                                                                                                                                                                                     | 0                                                          | 6               |  |      |
|                 |                                                                                                                                                                                                                                                                     | 0.2                                                        | 46              |  |      |
|                 |                                                                                                                                                                                                                                                                     | 0.4                                                        | 70              |  |      |
|                 |                                                                                                                                                                                                                                                                     | 0.8                                                        | 23              |  |      |
|                 |                                                                                                                                                                                                                                                                     | 1.6                                                        | 0               |  |      |
|                 |                                                                                                                                                                                                                                                                     | Allow ± 0.5 if they choose to quote germination % to 1 dp. |                 |  |      |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Additional Guidance                   | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------|
| 1(d)(ii)        | <p>An answer that includes the following points :</p> <ul style="list-style-type: none"> <li>• both (appear to) have optimum <u>at 0.4 mol dm<sup>-3</sup></u> (1)</li> <li>• (but for either) it could be anywhere between above 0.2 mol dm<sup>-3</sup> and below 0.8 mol dm<sup>-3</sup> (1)</li> <li>• values for optimum could be different (to each other) / they might not have the same optima / one optimum might not be 0.4 mol dm<sup>-3</sup> (1)</li> <li>• more concentrations of sucrose (between 0.2 and 0.8 mol dm<sup>-3</sup>) should be investigated (1)</li> </ul> | ACCEPT 0.2 – 0.8 mol dm <sup>-3</sup> | (3)  |